Enhanced-dissipation tools

Wiki *Enhanced dissipation for passive scalars*

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Contributors: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic "Enhanced-dissipation tools".

Tools specific to enhanced dissipation. A statement comes to the Toolkit when several proofs use it, or when it holds in more generality than one result; a step used by one proof only is a claim inside that proof. Each statement is given once, with its proof or a reference, and the proofs elsewhere link to it.

1 Tools

[For instance spectral-gap and resolvent lemmas, and the link between mixing and enhanced dissipation.]

1.1 From resolvent bounds to semigroup bounds

Definition 1 (m-accretive operator)**.** A closed operator H on a complex Hilbert space X is m-accretive if, for every $\mu \in \mathbb{C}$ with $\operatorname{Re}\mu > 0$, the operator $H + \mu$ is invertible and $\|(H+\mu)^{-1}\| \le (\operatorname{Re}\mu)^{-1}$. Then $D(H)$ is dense, $\operatorname{Re}\langle Hf, f\rangle \ge 0$ for $f \in D(H)$, and $-H$ generates a semigroup of contractions e^{-tH}.

Theorem 2 (Gearhart–Prüss type bound, Wei)**.** *Let H be an m-accretive operator on X, and let*

$$\Psi(H) := \inf\{\|(H - i\lambda)f\| \ : \ f \in D(H),\ \|f\| = 1,\ \lambda \in \mathbb{R}\}.$$

Then $\|e^{-tH}\| \le e^{\pi/2}\, e^{-\Psi(H)t}$ for all $t \ge 0$.

Idea. Cut the solution in time: $\chi(s)e^{-sH}f$ solves the equation with source $\chi'(s)e^{-sH}f$. After a Fourier transform in time the resolvent on the imaginary axis appears, and by Plancherel the definition of Ψ becomes a weighted inequality for the nonincreasing function $g(s) = \|e^{-sH}f\|^2$. With χ exponential in the bulk this forces g to decay like $e^{-2\Psi s}$. Bringing χ to zero at the two ends costs the constant $e^{\pi/2}$.

Proof. Sources. Wei (2021), Section 2, partly motivated by Helffer and Sjöstrand (2010).

Let $\Psi = \Psi(H)$. Since $D(H)$ is dense, it is enough to prove $\|e^{-tH}f\| \le e^{\pi/2 - \Psi t}\|f\|$ for $f \in D(H)$. Fix such an f and let $g(s) = \|e^{-sH}f\|^2$, which is nonincreasing because the semigroup is a contraction. In particular there is nothing to prove if $\Psi t \le \pi/2$, so we assume $\Psi t > \pi/2$.

Step 1 (an inequality for g).

Claim. *For every $l > 0$ and every real $\chi \in C^1([0,l])$ with $\chi(0) = \chi(l) = 0$,*

$$\int_0^l \chi'(s)^2 g(s)\, ds \ge \Psi^2 \int_0^l \chi(s)^2 g(s)\, ds.$$

Proof of the claim. Let $u(s) = \chi(s)e^{-sH}f$ and $w(s) = \chi'(s)e^{-sH}f$ on $[0,l]$, so that $\partial_s u + Hu = w$. For $\lambda \in \mathbb{R}$ let $\hat{u}(\lambda) = \int_0^l u(s)e^{-i\lambda s}\,ds$, and similarly $\hat{w}$. Since H is closed, $\hat{u}(\lambda) \in D(H)$. Integrating by parts in s, the boundary terms vanish because $\chi(0) = \chi(l) = 0$, and

$$(H + i\lambda)\hat{u}(\lambda) = \hat{w}(\lambda).$$

By the definition of Ψ, $\|\hat{w}(\lambda)\| \geq \Psi\|\hat{u}(\lambda)\|$ for every λ. By Plancherel's theorem for functions with values in X,

$$\int_0^l \|w(s)\|^2\,ds = \frac{1}{2\pi}\int_{\mathbb{R}} \|\hat{w}(\lambda)\|^2\,d\lambda \geq \frac{\Psi^2}{2\pi}\int_{\mathbb{R}} \|\hat{u}(\lambda)\|^2\,d\lambda = \Psi^2\int_0^l \|u(s)\|^2\,ds.$$

□

Step 2 (choice of χ). Let

$$t_1 = \frac{\pi}{4\Psi}, \qquad t_2 = t - \frac{\pi}{4\Psi}, \qquad t_3 = t + \frac{\pi}{4\Psi}, \qquad l = t + \frac{\pi}{2\Psi},$$

so that $t_1 < t_2$, and

$$\chi(s) = \begin{cases} \sin(\Psi s), & 0 \leq s \leq t_1, \\ \frac{1}{\sqrt{2}}\,e^{\Psi s - \pi/4}, & t_1 \leq s \leq t_2, \\ e^{\Psi t - \pi/2}\cos(\Psi(s-t)), & t_2 \leq s \leq l. \end{cases}$$

This function is C^1 and vanishes at 0 and l. Setting $h := \chi'^2 - \Psi^2\chi^2$, the claim reads $\int_0^l h\,g\,ds \geq 0$, and

$$h(s) = \begin{cases} \Psi^2\cos(2\Psi s), & 0 \leq s \leq t_1, \\ 0, & t_1 \leq s \leq t_2, \\ -\Psi^2 e^{2\Psi t - \pi}\cos(2\Psi(s-t)), & t_2 \leq s \leq l. \end{cases}$$

Hence $h \geq 0$ on $[0,t_1]$ and on $[t_3,l]$, $h \leq 0$ on $[t_2,t_3]$, and

$$\int_0^{t_1} h\,ds = \frac{\Psi}{2}, \qquad \int_{t_2}^{t} h\,ds = -\frac{\Psi}{2}e^{2\Psi t - \pi}, \qquad \int_t^l h\,ds = 0.$$

Step 3 (conclusion). Since g is nonincreasing, $hg \leq h\,g(0)$ on $[0,t_1]$, $hg \leq h\,g(t)$ on $[t_2,t]$, and $hg \leq h\,g(t_3)$ on $[t,l]$. Therefore

$$0 \leq \int_0^l h\,g\,ds \leq \frac{\Psi}{2}\,g(0) - \frac{\Psi}{2}\,e^{2\Psi t - \pi}g(t),$$

that is $g(t) \leq e^{\pi - 2\Psi t}g(0)$, which is the bound. □

References

B. Helffer and J. Sjöstrand, *From resolvent bounds to semigroup bounds*, preprint (2010), arXiv:1001.4171.

D. Wei, *Diffusion and mixing in fluid flow via the resolvent estimate*, Science China Mathematics **64** (2021), 507–518, arXiv:1811.11904.